E13 - Switching Phenomena

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Abstract

This experiment explores the dynamic voltage response of a capacitor in a MOSFET-switched circuit, with a focus on modeling the charging and discharging phases using appropriate differential equations. During the charging interval, the capacitor behavior was well-described by a first-order RC model. The fitted time constant of the initial measurement data was determined to be $\tau = (1.189 \pm 0.00139), \mu$s, closely matching the theoretical value derived from the known $R_{\text{charge}} = 1000, \Omega$ and the nominal $C = 1$ nF, τ was shown to increase as R_{charge} increases matching theoretical understanding. The fitted capacitance, $C = (1.19 \pm 0.06)$ nF, was slightly higher than the physical value, which is attributed to additional gate and layout capacitance in the circuit. In contrast, the discharge interval required a second-order underdamped RLC model to capture the observed oscillatory behavior. While the model reproduced the waveform's shape for each configuration, the extracted parameters—R, L, and C—deviated significantly from expectations, with fitted values such as $R = (26.2 \pm 3.59 \times 10^6), \Omega$ and $C = (0.42 \pm 57672), \text{nF}$ being physically implausible. These discrepancies are attributed to the complex nature of the discharge model along with physical phenomenon such as parasitic inductance, uncertain wire resistance, and non-ideal MOSFET characteristics. Despite high r^2 values in the fitting, the discharge model proved unreliable for quantitative parameter extraction. Overall, the experiment demonstrates the effectiveness of simple RC models for predicting capacitor charging behavior, while also exposing the challenges of modeling real-world high-speed switching events where parasitic effects dominate.

1 Initial Analysis
(Task I)

1.1 Initial Measurement Setup

Initial measurement of the MOSFET-switch circuit, was made with use of a 2000A series Picoscope to measure the voltage of the circuit, an Arduino Nano to apply a controlled 5V source to the gate of the IRF520 MOSFET as well as a constant 3.3V to the circuit. The circuit was constructed as seen in Fig 1. The R_{charge} was chosen to be 100Ω and capacitor $C = 1$ nF, replacing the $R_{\text{discharge}}$ was a solid-core jumper wire of resistance R_{wire}.

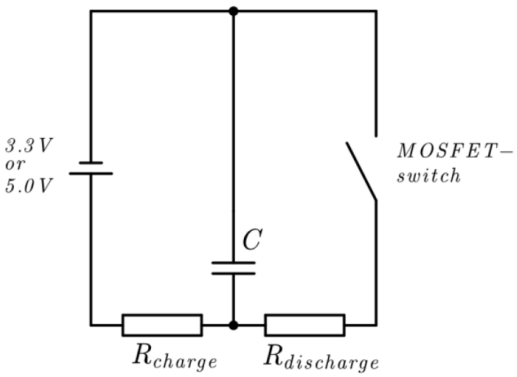

Figure 1: MOSFET-Switch Circuit Diagram

Source: [4]

Using the Arduino Nano, the MOSFET was setup to periodically open and close, and through use of the Picoscope the voltage over time was measured using a "repeat-rising" trigger set to 2.0V to collect the initial charge interval for a total of twelve waveforms, $n = 12$, and the mean taken as the initial measurement to increase precision. Then the trigger was set to "repeat-failing" at 2.0V to collect the discharge interval for a total of twelve waveforms, again taking the mean measurement for precision.

1.2 Initial Measurement Analysis

A multimeter was used to measure $R_{\text{wire}} \approx 10\ \Omega$, however the reading changed rapidly and wildly from $\approx 1\Omega$ to $\approx 30\Omega$. It is unknown as to why this was the case, but as such the accuracy and precision of this measurement is questionable. It was determined that the two intervals

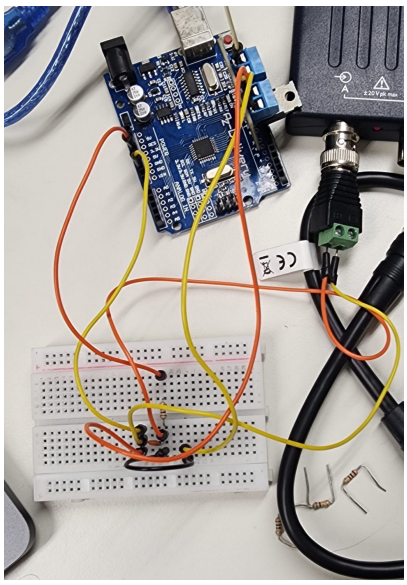

Figure 2: Initial MOSFET-Switch Circuit Diagram

of interest from the raw data of the voltage across time, was the charge interval of the capacitor immediately after the MOSFET switches on and the discharge interval immediately after the MOSFET switches off. These voltage trends are shown in Fig.??. The points at which the MOS-

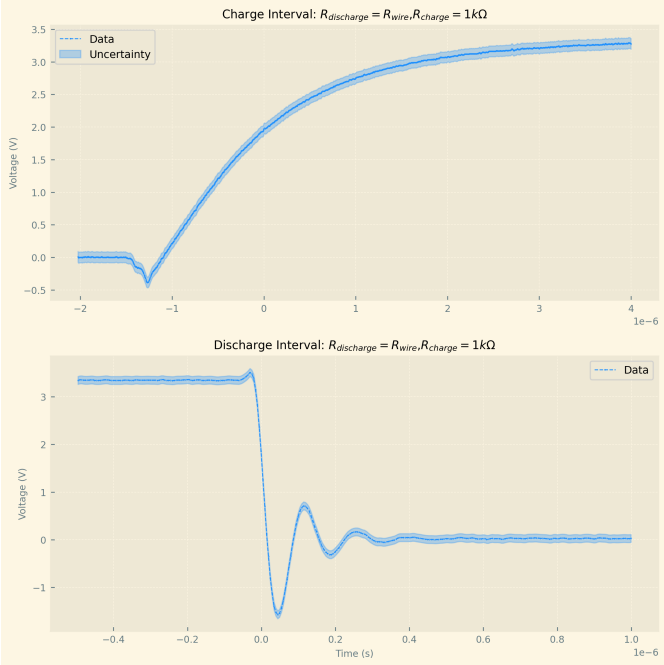

Figure 3: Initial Voltage Data showing both the charge and discharge intervals

FET switches, for both the charge and discharge intervals are of interests as there is a voltage drop and voltage increase respectively.
It is expected that the voltage drop for the charge interval is a result of the MOSFET's characteristics and factors such as gate-capacitance and internal "on-resistance", along with the instantaneous surge of electricity when the circuit is close. However it is unknown as to why, or how this value dropped below zero volts which indicates a generated reverse electromotive force. It is hypothesized that even during the charging phase, when the MOSFET first switches on, the sudden connection to a voltage source allows a very rapid inrush current to flow into the capacitor. If the path includes any inductance (parasitic inductance in wires, traces, or the MOSFET package), the abrupt change in current di/dt will induce a voltage. This can result in a brief negative voltage spike at the measuring point. After this drop the voltage follows a typical raise in charging trend as outlined in literature [2] of:

$$V = V_0(1 - e^{-\frac{t}{\tau}}) : \tau = R_{\text{charge}}C \tag{1}$$

The voltage trend of the discharge interval however, is significantly different to the expected discharge of a capacitor, as outlined in literature values [4]:

$$V = V_0 e^{-\frac{t}{\tau}} : \tau = R_{\text{discharge}}C \tag{2}$$

At the point at which the MOSFET switches off at $t \approx -40$ nm there is a voltage increase which is expected to be a result of both the MOSFET's internal "on-resistance" being removed from the circuit and the gate capacitance discharging at the instant the switch turns off. After this instantaneous increase in voltage the voltage drops but, unexpectedly undergoes a prominent under-damped, decaying oscillation until reaching $\approx 0V$, (varies due to uncertainty and voltage leakage). This trend is reminiscent of an RLC-Circuit switching response as shown below,

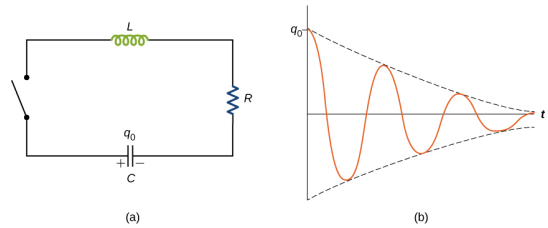

Figure 4: RLC-Switching Discharge response
Source: [3]

which is modeled with the following equation for the underdamped case ($\zeta < 1$):

$$V(t) = V_0\, e^{-\zeta\omega_0(t-t_0)} \cos(\omega_d(t - t_0)) \tag{3}$$

Where:

$\omega_0 = \frac{1}{\sqrt{LC}}$: is the undamped natural frequency,
$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$: is the damping ratio,
$\omega_d = \omega_0\sqrt{1 - \zeta^2}$: is the damped angular frequency,
t_0 : is the time when the MOSFET switch closes.

Eq.(3), implies an inductance in the circuit, it is hypothesized that this inductance is a generated inductance due to the presence of the MOSFET and the fast influx of current, causing a parasitic inductance within the circuit and creating the RC-Switching circuit to act as a momentary RLC-Switching circuit.

2 Model Construction
(Task II)

From analysis of the initial data, The constructed model, only models the area of interest, this being the charge and discharge intervals. For the charge interval Eq.(2) was used and for the discharge interval Eq.(3). was used, as it is believed that is RLC-Switching equation will more accurately describe the voltage trend observed despite the circuit being a RC-switching circuit. Thus the model used is:

$$V(t) = \begin{cases} V_0\left(1 - e^{-t/(R_{\text{charge}}C)}\right) & \text{for } t < t_0 \\ V_0 e^{-\zeta\omega_0(t-t_0)} \cos(\omega_d(t - t_0)) & \text{for } t \geq t_0 \end{cases} \tag{4}$$

To understand the voltage evolution across the capacitor during switching, we divide the system's behavior into two distinct phases: charging and discharging. Each phase is governed by its own physical model and corresponding differential equation.

2.1 Charging Phase: First-Order RC Charging Model

When the MOSFET is off, the capacitor charges through a resistor R_{charge}. The charging current flows from a DC voltage source (V_0) through this resistor into the capacitor. This process is governed by a first-order differential equation:

$$\frac{dV(t)}{dt} = \frac{1}{R_{\text{charge}}C}\left(V_0 - V(t)\right)$$

The solution to this equation gives the voltage across the capacitor as a function of time:

$$V_{\text{charge}}(t) = V_0 \left(1 - e^{-t/(R_{\text{charge}}C)}\right)$$

This equation represents an exponential rise in voltage from 0 towards the supply voltage V_0. The rate of this rise is determined by the time constant $\tau = R_{\text{charge}}C$, which characterizes how quickly the capacitor charges. The voltage asymptotically approaches V_0 as $t \to \infty$.

2.2 Discharging Phase: Second-Order RLC Model

When the MOSFET is turned on at time $t = t_0$, the capacitor discharges through a very small resistance $R_{\text{discharge}}$. However, the discharge path also includes parasitic inductance L from wiring and component leads, which cannot be ignored due to the fast switching and high-frequency nature of the transient response.
This leads to a second-order differential equation describing the voltage across the capacitor:

$$L\frac{d^2V}{dt^2} + R_{\text{discharge}}\frac{dV}{dt} + \frac{1}{C}V = 0$$

This equation has three terms, each representing a different physical mechanism:

- $L\frac{d^2V}{dt^2}$: accounts for inertial effects due to inductance, resisting changes in current.
- $R\frac{dV}{dt}$: models resistive damping, which dissipates energy and causes the oscillation to decay.
- $\frac{1}{C}V$: corresponds to the restoring force from the capacitor's electric field.

Together, these terms describe a damped harmonic oscillator, electrically analogous to a mass-spring-damper system.
For the underdamped case (which matches our experimental observation of ringing), the general solution is:

$$V_{\text{discharge}}(t) = V_0\, e^{-\zeta\omega_0(t-t_0)} \cos\left(\omega_d(t - t_0) + \phi\right)$$

where:

- $\omega_0 = \dfrac{1}{\sqrt{LC}}$ is the undamped natural angular frequency
- $\zeta = \dfrac{R_{\text{discharge}}}{2}\sqrt{\dfrac{C}{L}}$ is the damping ratio
- $\omega_d = \omega_0\sqrt{1-\zeta^2}$ is the damped angular frequency
- V_0 and ϕ are constants determined by the initial voltage and current at $t = t_0$

This solution describes an oscillation that decays over time. The oscillation arises due to energy exchange between the inductor and capacitor. The decay (damping) is caused by energy loss through the resistor. The frequency and shape of the waveform match the observed ringing in the voltage signal .

2.3 Proof of Discharge Voltage Equation

Since the exact equation for the charging voltage is provided in the cited source[1] and only the differential equation is provided for the discharge phase of the circuit, to make make sure that the Eq.(3) holds, a proper derivation is provided below.
It is given that the second-order linear differential equation for an RLC series circuit (during discharge), as outlined in literature [] is:

$$L\frac{d^2V}{dt^2} + R\frac{dV}{dt} + \frac{1}{C}V = 0$$

It is needed to verify that the underdamped solution:

$$V(t) = V_0\, e^{-\zeta\omega_0 t}\cos(\omega_d t)$$

is indeed a solution of the above equation,
Let's propose:

$$V(t) = V_0 e^{-\zeta\omega_0 t}\cos(\omega_d t)$$

Computing its first derivative:

$$\frac{dV}{dt} = V_0\left[-\zeta\omega_0 e^{-\zeta\omega_0 t}\cos(\omega_d t) - \omega_d e^{-\zeta\omega_0 t}\sin(\omega_d t)\right]$$

Now the second derivative:

$$\begin{aligned}\tfrac{d^2V}{dt^2} = V_0\,&\left[(\zeta^2\omega_0^2 - \omega_d^2)e^{-\zeta\omega_0 t}\cos(\omega_d t)\right.\\ &+2\zeta\omega_0\omega_d e^{-\zeta\omega_0 t}\sin(\omega_d t)\end{aligned}$$

Substituting $V(t)$, $\frac{dV}{dt}$, and $\frac{d^2V}{dt^2}$ into:

$$L\frac{d^2V}{dt^2} + R\frac{dV}{dt} + \frac{1}{C}V$$

Grouping terms for $\cos(\omega_d t)$ and $\sin(\omega_d t)$ separately:
Cosine term:

$$\begin{aligned}LV_0(\zeta^2\omega_0^2 - \omega_d^2)e^{-\zeta\omega_0 t}\cos(\omega_d t) + R\left[-\zeta\omega_0 V_0 e^{-\zeta\omega_0 t}\cos(\omega_d t)\right]\\ +\tfrac{1}{C}V_0 e^{-\zeta\omega_0 t}\cos(\omega_d t)\end{aligned}$$

Factor out $V_0 e^{-\zeta\omega_0 t}\cos(\omega_d t)$:

$$V_0 e^{-\zeta\omega_0 t}\cos(\omega_d t)\left[L(\zeta^2\omega_0^2 - \omega_d^2) - R\zeta\omega_0 + \frac{1}{C}\right]$$

Sine term:

$$L\cdot 2\zeta\omega_0\omega_d V_0 e^{-\zeta\omega_0 t}\sin(\omega_d t) + R\left[-\omega_d V_0 e^{-\zeta\omega_0 t}\sin(\omega_d t)\right]$$

Factor out $V_0 e^{-\zeta\omega_0 t}\sin(\omega_d t)$:

$$V_0 e^{-\zeta\omega_0 t}\sin(\omega_d t)\left[2L\zeta\omega_0\omega_d - R\omega_d\right]$$

To prove the solution is correct, both bracketed expressions must equal zero.
For the sine term:

$$2L\zeta\omega_0\omega_d = R\omega_d \quad\Rightarrow\quad \zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$$

For the cosine term:
Using:

- $\omega_0 = \frac{1}{\sqrt{LC}}$
- $\omega_d^2 = \omega_0^2(1-\zeta^2)$

$$L(\zeta^2\omega_0^2-\omega_d^2)-R\zeta\omega_0+\frac{1}{C} = L(\zeta^2\omega_0^2-\omega_0^2+\zeta^2\omega_0^2)-R\zeta\omega_0+\frac{1}{C}$$

Simplifies to:

$$L(-\omega_0^2+2\zeta^2\omega_0^2)-R\zeta\omega_0+\frac{1}{C} = -L\omega_0^2+2L\zeta^2\omega_0^2-R\zeta\omega_0+\frac{1}{C}$$

Substitute $L\omega_0^2 = \frac{1}{C}$:

$$-\frac{1}{C}+2L\zeta^2\omega_0^2-R\zeta\omega_0+\frac{1}{C}=0$$

This shows that the equation:

$$V(t) = V_0 e^{-\zeta\omega_0 t}\cos(\omega_d t)$$

is exactly a solution of the second-order RLC differential equation when the damping ratio ζ and frequencies are defined as above. For simplicity this equation was proved as a solution, however in the general case the solution must include a phase shift as given in Eq.(3) since the general solution will be a linear combination of sine and cosine:

$$V(t) = e^{-\zeta\omega_0 t}(A\cos(\omega_d t)+B\sin(\omega_d t))$$

2.4 Application of Model to Initial Measurement

From Eq.(4). with multiple free parameters for both the charge interval and the discharge interval. For the charge interval, the free parameters were determined to be τ,V_0 and C. C was chosen to be a free parameter as it is believed that the gate capacitance will significantly alter the voltage trend and thus C was left free to account for this additional capacitance. For the discharge interval all parameters: V_0, L, C, R and t_0 were left free as it is expected that L, C and R, will be significantly different from obvious circuit component values due to the parasitic inductance, internal resistance and the gate capacitance of the MOSFET.

Fig.5. shows the fitting of the developed model on the initial measurement data. The time domain of both charge and discharge interval was zeroed to the point at which the MOSFET switches on or off respectively. The results show a very strong correlation between experimental and fitted data with $r^2 = 0.9996$ and $r^2 = 0.9979$ for the charge and discharge intervals respectively. The determined RC constant τ for the charge interval was determined to be $\tau = (1.1389 \pm 1.39\cdot 10^{-3})\mu$s which is of the correct order of magnitude. A rough estimate is that $6\tau \approx$ the time for the voltage to reach its maximum, $V = 3.3V$, the determined τ roughly agrees with this estimate. Additionally the results determined $C = (1.19 \pm 0.0595)$ nF, is higher than the physical minimum of $C \geq 1$ nF due to the capacitor in the circuit, Fig.3., it is expected that the additional capacitance in the determined value arises from the MOSFET's gate capacitance.

The discharge interval, shows a strong correlation however for all parameters, except V_0 and t_0 there is a very large uncertainty, although due to the nature of the discharge interval equation in Eq.(4). the uncertainty in the fit is

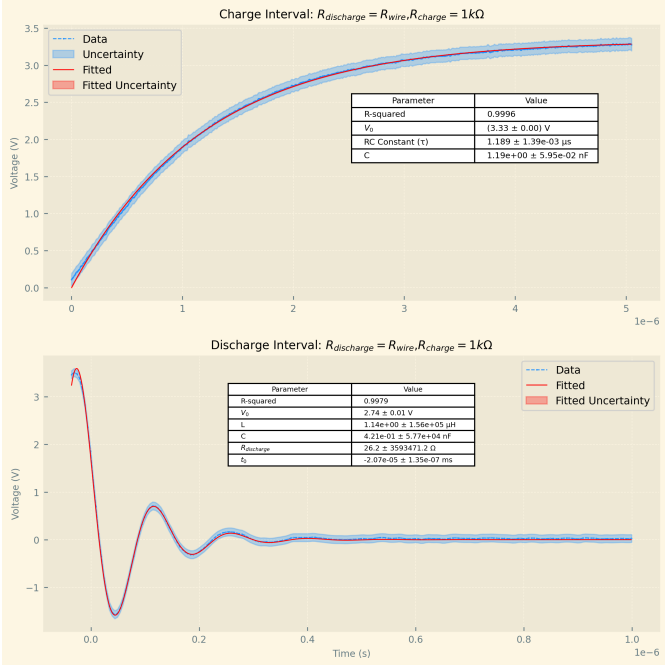

Figure 5: Developed Model applied to the charge and discharge intervals of the initial data

significantly lower than intuition suggests. Although the $R_{\text{discharge}} = (26.2 \pm 3593471.2)\Omega$ is significantly above the expected $R_{\text{wire}} \approx 10\Omega$ it is expected that the total R_{total} is larger due to unaccounted for resistances in components and the internal resistance of the MOSFET. The determined capacitance C is wildly off the expected value, it is expected that that capacitance is $C \geq 1$ nF, however below 1 nF is not physically possible, however it is also of note that the significantly large uncertainty of this parameter suggests little confidence in the determined value of C.

The discharge model used also determined a generated inductance $L = (1.14 \pm 1.56 \cdot 10^5)\mu$H. It was expected that due to the fast switching off time of the MOSFET an inductance is momentarily generated and this model suggests that to be the. However again, very large uncertainty in any associated parameter significantly reduces the confidence of its value.

3 Further Analysis
(Task III)

3.1 Measurement Setup

To further investigate the validity of the model developed, Eq.(4). additional measurements using the setup and methodology as outlined in Sec.1. was collected using a range of Resistors, $R_{\text{discharge}} \in [R_{\text{wire}}, 1\Omega, 10\Omega, 100\Omega, 1000\Omega]$ and $R_{\text{charge}} \in [1000\Omega, 2200\Omega, 10\text{k}\Omega]$ to assess the validity of the model developed with different values of R_{charge} and $R_{\text{discharge}}$ as well as determined the effect that changing these resistances has on the voltage trend.

3.2 Measurement Analysis

Fig.6. and Fig.7 are two of the total of 7 different circuit configurations that were measured, that exhibit trends

that are significantly different from the initial measurement trends. Fig.6 shows that at a high $R_{\text{charge}} = 10\text{k}\Omega$, $\tau = (11.831 \pm 0.00102)\mu\text{s}$ which is significantly larger than that of the $R_{\text{charge}} = 1000\Omega$ which was consistency on the order of $1\mu\text{s}$, along with a fitted $C = (1.18 \pm 0.0592)\text{nF}$ which is a physically possible value $\geq 1\text{nF}$ in contrast to the result of the initial measurement fitting.

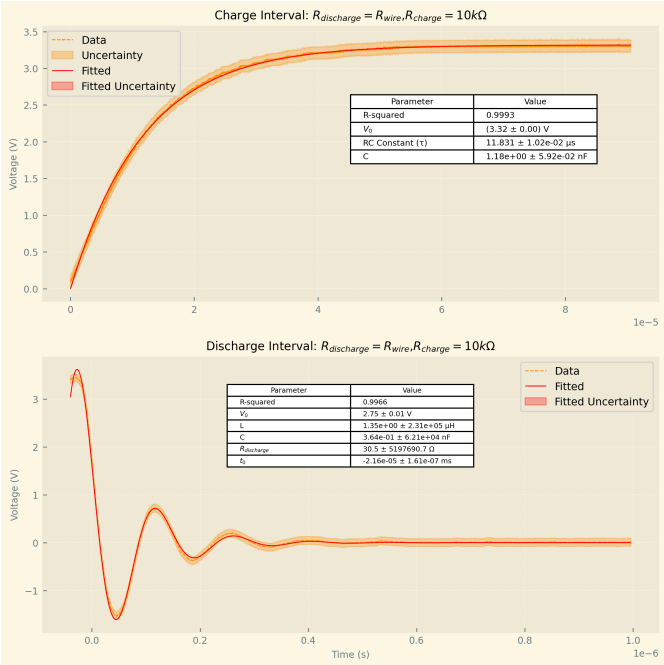

Figure 6: Fitted Charge and Discharge intervals of a $R_{\text{discharge}} = R_{\text{wire}}$ and $R_{\text{charge}} = 10\text{k}\Omega$ configuration

Fig.7. shows that the discharge model failed to find a valid solution for the measurement data. The figure depicts, what is believed to be a critically damped trend, the model developed is for the underdamped case of oscillating exponential decay. As such the model was unable to model the critically damped case, that was seen for $R_{\text{discharge}} \geq 100\Omega$, additionally this case is a limitation to the model.

Across all configurations the discharge model display very large uncertainties, on average the relative uncertainty for the charge model was $\approx 5.08\%$ while the relative uncertainty in the discharge model was a staggering $\approx 25059730.59\%$ suggesting little precision nor confidence in the determined parameters of the discharge model across all measurement configurations.

3.3 τ and C Analysis
(Task IV)

The RC constant τ was only present for the charge interval and associated model of the voltage model equation Eq.4. Fig.8. shows the determined τ for each configuration. From the overall trend and parameters across each configuration, *(See Appendix 1.)*, τ appears to be largely dependent on the capacitor, C and R_{charge} resistor and minimally dependent on $R_{\text{discharge}}$. this dependence follows a trend that increasing the R_{charge} increases τ which is in agreement with both literature understanding and the model used. Fig.8. supports this as all configurations $R_{\text{charge}} = 1000\Omega$ have comparable τ values, where as $R_{\text{charge}} = 10\text{k}\Omega$ have significantly different τ values. $R_{\text{charge}} = 2.2\text{k}\Omega$ is considered an outlier to this trend as

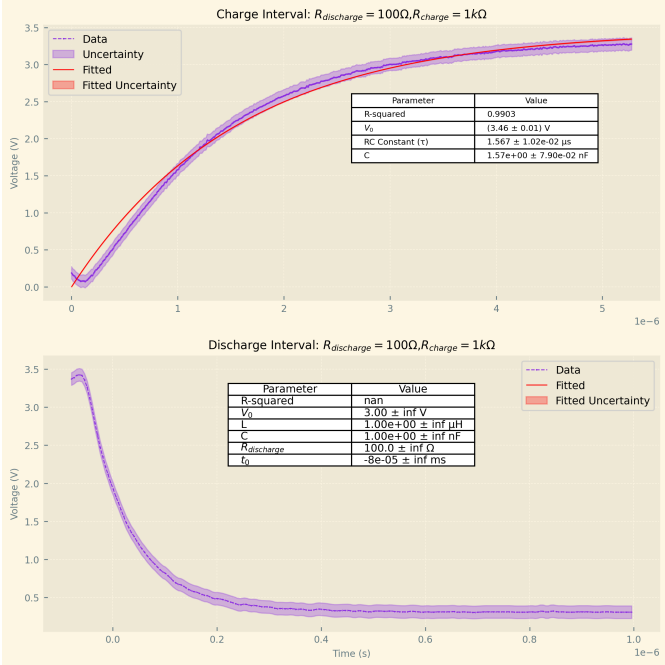

Figure 7: Fitted Charge and Discharge intervals of a $R_{\text{discharge}} = 100\Omega$ and $R_{\text{charge}} = 1\text{k}\Omega$ configuration

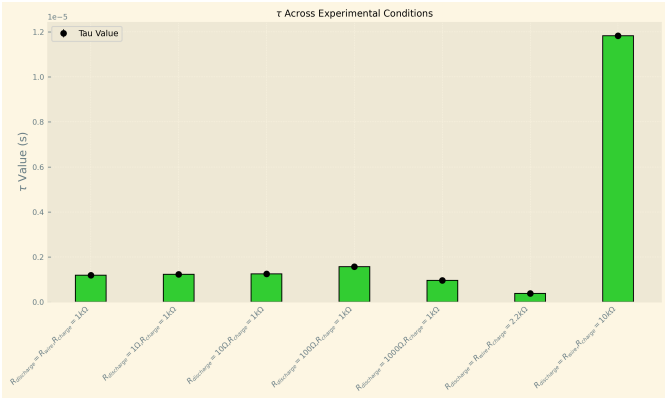

Figure 8:

its value is in opposition to the model and physical understanding of the processes present.

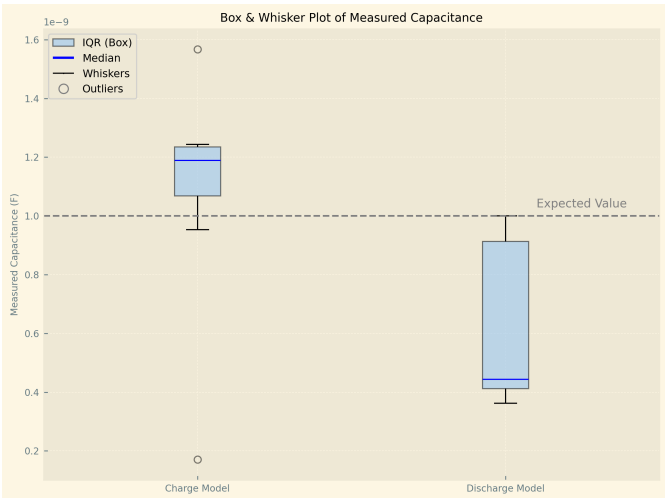

Figure 9: Box & Whisker Plot for the

When assessing the differing C parameters determined across all configurations Fig.9. shows a Box & Whisker

plot of the determined capacitance values. The graph conveys that the discharge model is imprecise and possess large deviation across configurations when it is expected that they be largely similar. The capacitance values determined through the charge model are physically valid as most lie above the minimum of $C \geq 1$ nF, and aligns with expectations of a higher C, as a result of gate capacitance by the MOSFET.

Across all configurations the Voltage model Eq.4. models the voltage trend however fails to determine valid parameters for its fitting. The discharge model fails to relate determined parameters to physical components and values, whilst maintaining a staggeringly high uncertainty with an average relative uncertainty in parameters of $\approx 25059730.59\%$. The charge model is more accurate and follows an increasing τ trend as R_{charge} increases, *(with exception of $R_{charge} = 2.2k\Omega$)*, and its determined values for C are physically valid, as most lie above the minimum of $C \geq 1$ nF. It is predicted that the minor shortcomings of the charge model, in terms of accuracy, is a result of the decision to make that capacitance C a free variable, and the shortcomings of the discharge model is a result of the complicated inter-related nature of ζ, ω_0 and ω_d with physical parameters, R,L and C.

4 Discussion

The modeling and fitting process in this experiment provided valuable insights into the physical behavior of the capacitor-MOSFET circuit, while also highlighting several discrepancies between the actual component values and those obtained through curve fitting—particularly in the discharge interval. In the charging phase, the extracted parameters aligned well with expectations. The applied voltage was known to be 3.3 V and the capacitor used had a value of 1 nF. The fitted values—such as a time constant $\tau = (1.189 \pm 0.00139)\mu s$ and a capacitance $C = (1.19 \pm 0.06)$nF—are consistent with these known values and confirm the validity of the first-order exponential model. The slightly elevated capacitance can reasonably be attributed to additional parasitic capacitance introduced by the MOSFET gate and circuit layout. This interpretation is reinforced by the smooth exponential rise of the voltage, the high correlation coefficient ($r^2 \approx 0.9996$), and the physically reasonable time constant, which follows the expected relation $\tau = R_{charge} \times C$.

In contrast, the discharging phase presented significant inconsistencies. Although the waveform clearly displayed underdamped oscillations that qualitatively match an RLC-type response, the fitted parameters were extremely unstable and often physically implausible. The fitted discharge resistance, for example, reached absurd values with enormous uncertainty—$R = (26.23.59 \times 10^6)\Omega$ in one case—despite the actual resistance being a wire of approximately 10 Ω. Likewise, the fitted capacitance frequently dropped below 1 nF, which is not physically possible given the known capacitor in the circuit, and the inductance values were similarly distorted with uncertainties on the order of $10^5 \mu H$. These results suggest that while the model is mathematically flexible enough to reproduce the general shape of the discharge waveform, it fails to yield meaningful component values.

This discrepancy likely stems from a combination of modeling limitations and physical factors. The discharge behavior includes parasitic inductance from the wiring and PCB traces, which varies sensitively with layout and is difficult to quantify. The MOSFET itself introduces nonlinear and dynamic behavior, such as gate capacitance effects, temperature-dependent resistance, and transient switching artifacts that are not captured by a linear RLC model. Furthermore, the discharge signal is typically brief and oscillatory, making it highly sensitive to noise and difficult to analyze with stable curve-fitting. The interdependence of model parameters such as R, L, C, t_0, and ω exacerbates the instability, allowing many combinations of parameters to fit the data comparably well, but without physical meaning.

It is also important to recognize that the mathematical model used assumes idealized linear elements, which do not hold in practice. In particular, the transition from underdamped to critically damped behavior—as seen in high-resistance discharge configurations—is not handled by the model, which was limited to underdamped forms. This renders the model incapable of describing certain measurements, as demonstrated by its failure in configurations with $R_{discharge} \geq 100\Omega$.

Despite these limitations, the model succeeded in qualitatively confirming the presence of parasitic inductance and a second-order dynamic response. However, its use as a quantitative fitting tool must be approached with caution. The discharge model produces acceptable fits visually, but the extracted parameters are unreliable. By contrast, the charge model remains consistent across configurations, with clear trends showing increased τ with increased R_charge, and capacitance values that stay near or above the expected minimum of 1 nF.

Improving the accuracy of the discharge analysis would require using better-defined resistors, minimizing trace inductance, and possibly introducing known inductors for calibration. The addition of snubber circuits could also suppress unwanted ringing. Furthermore, alternative analysis techniques such as direct extraction of damping ratio or frequency from the waveform, rather than fitting all parameters simultaneously, may yield more stable results.

4.1 Conclusion

This experiment successfully demonstrated and modeled the switching behavior of a capacitor in an electronic circuit involving a MOSFET switch. By separating the analysis into charging and discharging intervals, and applying appropriate mathematical models to each phase, the voltage behavior over time was effectively captured and interpreted.

For the charging interval, the first-order RC model provided a robust and accurate description of the system. The fitted parameters for time constant and capacitance were in close agreement with expected values and reflected physically meaningful results. This confirmed that the RC charging model is both reliable and predictive under

the experimental conditions used.

The discharging interval, on the other hand, revealed significant deviations between theoretical expectations and fitted parameters. Although the underdamped RLC model qualitatively reproduced the observed oscillatory behavior, it failed to yield quantitatively reliable values for resistance, inductance, and capacitance. This was largely due to the presence of parasitic inductance, poorly defined resistance in the jumper wire, non-ideal characteristics of the MOSFET, and the mathematical interdependence of the model parameters. Additionally, the model was limited to the underdamped case and could not accommodate critically damped scenarios observed with higher discharge resistances.
Despite these limitations, the experiment demonstrated that fast switching events in even simple RC circuits can introduce higher-order effects, such as parasitic inductance, leading to complex transient behavior. The high precision of the charging model and the qualitative success of the discharging model affirm the value of such modeling in understanding real-world electronic behavior, while also underscoring the importance of considering non-ideal effects when analyzing high-speed circuits.

Overall, the experiment reinforced theoretical concepts from electrodynamics while highlighting the limitations of ideal models in practical applications. For further refinements in experimental setup and modeling approaches it is recommended to reduce the discharge model parameters to improve the quantitative reliability and precision of discharge parameter extraction and offer deeper insights into transient circuit dynamics.

References

[1] W. Demtröder. *Electrodynamics and Optics*. Springer, 2nd edition, 2010.

[2] C. R. Nave. Capacitor discharge, 2024. URL http://hyperphysics.phy-astr.gsu.edu/hbase/electric/capdis.html. HyperPhysics, Georgia State University. Accessed 2025-05-26.

[3] OpenStax College. Rlc series circuits. https://openstax.org/books/university-physics-volume-2/pages/14-6-rlc-series-circuits, 2016. CC BY license. Accessed 2025-05-26.

[4] M. Ziese. *E13h Switching Phenomena*. Universität Leipzig, May 2025.

Appendix 1. Table of all determined parameters

$R_{\textbf{discharge}}$ (Ω)	$R_{\textbf{charge}}$ (Ω)	**Discharge** r^2	**Charge** r^2	**Charge** C **(nF)**	**RC Constant** τ (μs)	**Discharge** C **(nF)**	**Discharge** R (Ω)
R_{wire}	1000	0.9996	0.9979	1.19 ± 0.06	1.189 ± 0.00139	0.42 ± 57672.44	26.2 ± 3593471.2
1	1000	0.9996	0.9979	1.23 ± 0.06	1.226 ± 0.00132	0.60 ± 117249.19	13.7 ± 2669461.2
10	1000	0.9997	0.9938	1.24 ± 0.06	1.243 ± 0.00135	0.37 ± 113311.47	38.3 ± 11699885.5
100	1000	0.9903	N/A	1.57 ± 0.08	1.567 ± 0.0102	N/A	N/A
1000	1000	0.7113	N/A	0.95 ± 0.05	0.953 ± 0.0209	N/A	N/A
R_{wire}	2200	0.9863	0.9879	0.17 ± 0.01	0.375 ± 0.00236	0.36 ± 119915.10	33.9 ± 11308123.6
R_{wire}	10000	0.9993	0.9966	1.18 ± 0.06	11.831 ± 0.0102	0.36 ± 62146.61	30.5 ± 5197690.7